

Due Date: 2/21/2023

Group 1 MATH 341

Homework 2

Problem 1

a)

HW 2 2/1/23

$f(x) = \tan(x)$ $a = \frac{\pi}{4}$ $[\alpha, \beta] = [0, \pi/4]$

Compute Taylor Polynomials $p_n(x)$ for $n = 1, 2, 3$ about point a

$\star \sec^2(x) = \frac{1}{\cos^2(x)} = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = 1 + \frac{\sin^2(x)}{\cos^2(x)} = 1 + \tan^2(x)$

$f(x) = \tan(x)$

$f'(x) = 1 + \tan^2(x)$

$f''(x) = 2 \tan(x) (1 + \tan^2(x)) = 2 \tan(x) + 2 \tan^3(x)$

$f'''(x) = 2(1 + \tan^2(x)) + 6 \tan^2(x) (1 + \tan^2(x)) = 2 + 2 \tan^2(x) + 6 \tan^2(x) + 6 \tan^4(x)$
 $= 2 + 8 \tan^2(x) + 6 \tan^4(x)$

$f^{(4)}(x) = 16 \tan(x) (1 + \tan^2(x)) + 24 \tan^3(x) (1 + \tan^2(x))$
 $= 16 \tan(x) + 16 \tan^3(x) + 24 \tan^3(x) + 24 \tan^5(x)$
 $= 16 \tan(x) + 40 \tan^3(x) + 24 \tan^5(x)$

$f(a) = 1$
 $f'(a) = 1 + 1 = 2$
 $f''(a) = 2 + 2 = 4$
 $f'''(a) = 2 + 8 + 6 = 16$
 $f^{(4)}(a) = 16 + 40 + 24 = 80$

$p_1(x) = f(a) + f'(a)(x-a) \rightarrow = 1 + 2(x - \frac{\pi}{4}) = 1 + 2(x - \pi/4)$

$p_2(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 \rightarrow = 1 + 2(x - \pi/4) + \frac{4}{2}(x - \pi/4)^2 = 1 + 2(x - \pi/4) + 2(x - \pi/4)^2$

$p_3(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 \rightarrow = 1 + 2(x - \pi/4) + 2(x - \pi/4)^2 + \frac{16}{6}(x - \pi/4)^3 = 1 + 2(x - \pi/4) + 2(x - \pi/4)^2 + \frac{8}{3}(x - \pi/4)^3$

b)

We set x as a domain from 0 to $\pi/4$ taking $\pi/512$ steps and y as $\tan(x)$. After that, we set p_1 , p_2 , and p_3 with the answers we got in problem 1(a). We did two different approaches to get our graphs. In the Matlab script we plot the function using plot command with 'k' to can a solid black line. Then we turn on the grid of the graph with grid on, then label x,y, and title of the graph.

After that, we used the hold on command to then add on the plots of p_1 ('--b', using blue dash line), p_2 ('-.' using red dash dot line), and p_3 (':m', using pink quotation line) with a legend of $f(x)$, p_1 , p_2 , and p_3 located in the southeast of the graph.

In Octave script, instead of using hold on they put $f(x)$, p_1 , p_2 , and p_3 in one plot command (i.e, plot(x,y,'-k',x,p1,'--b',x,p2,'-.',x,p3,':m')), which does look neater since less code to look at. But it does the same thing as the Matlab script.

Here is the script for Matlab:

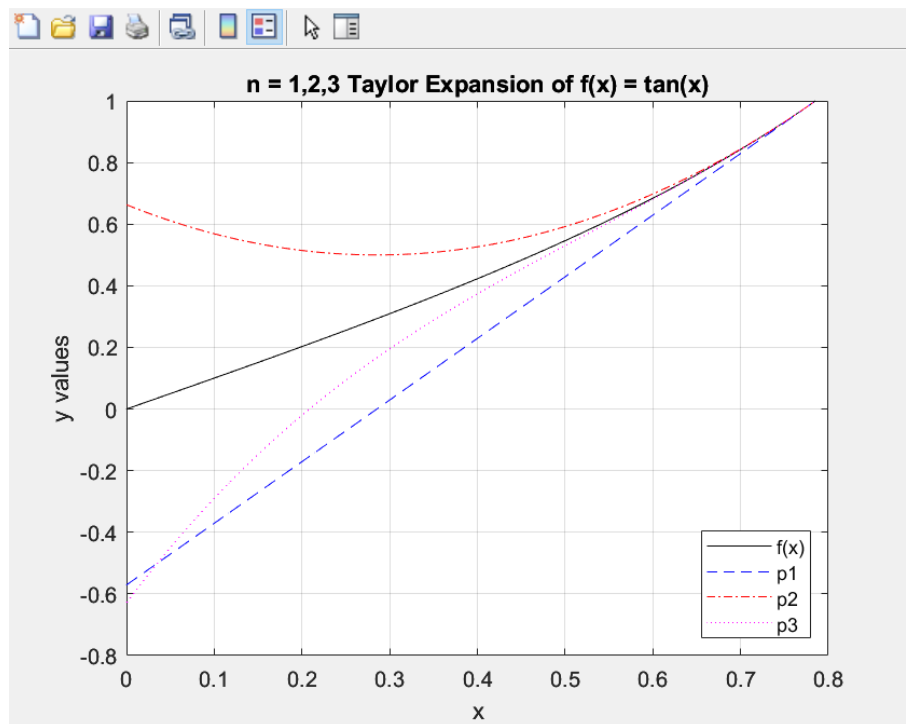
```
x = 0:pi/512:pi/4;
y = tan(x);
plot(x,y,'k')
grid on
xlabel('x');
ylabel('y values');
title('n = 1,2,3 Taylor Expansion of f(x) = tan(x)')
```

```

hold on
p1 = 1 + 2*(x-pi/4);
plot(x,p1, '--b')
hold on
p2 = 1 + 2*(x-pi/4) + 2*(x-pi/4).^2;
plot(x,p2, '-.r')
hold on
p3 = 1 + 2*(x-pi/4) + 2*(x-pi/4).^2 + 8/3*(x-pi/4).^3;
plot(x,p3, ':m')
legend ('f(x)', 'p1', 'p2', 'p3', 'Location', 'southeast')
hold off

```

Graph from matlab



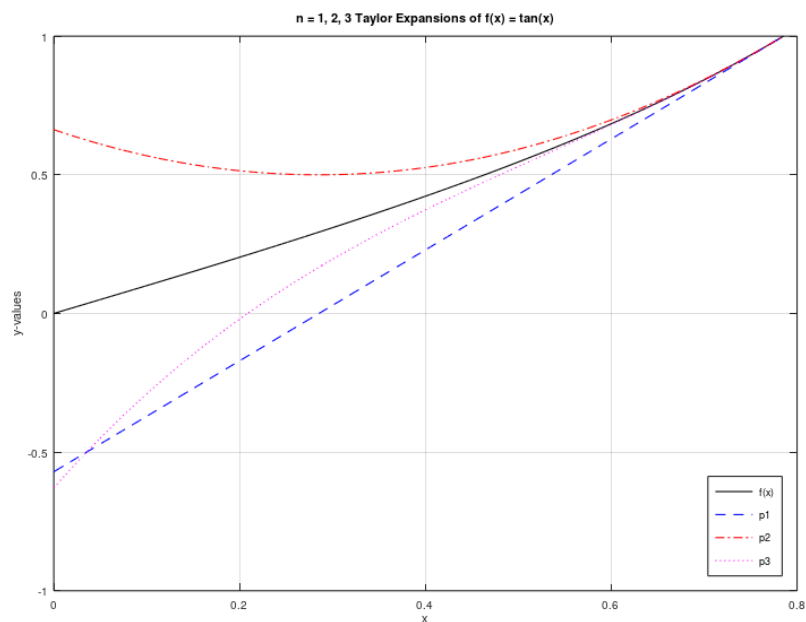
Octave Script:

```

x = 0:pi/512:pi/4 # (b-a)/128 -> (pi/4)/128 = pi/512
y = tan(x)
p1 = 1 + 2*(x-pi/4)
p2 = 1 + 2*(x-pi/4) + 2*(x-pi/4).^2 # component-wise power operator
p3 = 1 + 2*(x-pi/4) + 2*(x-pi/4).^2 + 8/3*(x-pi/4).^3
plot(x,y,'-k',x,p1,'--b',x,p2,'-.r',x,p3,':m');
grid on;
xlabel ('x');
ylabel ('y-values');
title('n = 1, 2, 3 Taylor Expansions of f(x) = tan(x)');
legend("f(x)", "p1", "p2", "p3", 'Location', 'southeast')

```

Graph from octave



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c)

use Remainder formula for Taylor Approximation to bound remainders $R_{n+1}(x)$ of $p_n(x)$ $n=1, 2, 3$

$$R_{n+1}(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi) (x-a)^{n+1}$$

$n=1 \rightarrow R_2(x) = \frac{1}{2!} f^{(2)}(\xi) (x-\frac{\pi}{4})^2 = \frac{1}{2} (2 \tan(\xi) + 2 \tan^3(\xi)) |x-\frac{\pi}{4}|^2$

$\tan(x)$ is monotone on $[0, \frac{\pi}{4}]$

bound ξ : $0 \leq \xi \leq \frac{\pi}{4} \rightarrow \tan(0) \leq \tan(\xi) \leq \tan(\frac{\pi}{4}) \rightarrow 0 \leq \tan(\xi) \leq 1$

bound x : $0 \leq x \leq \frac{\pi}{4} \rightarrow -\frac{\pi}{4} \leq x - \frac{\pi}{4} \leq 0 \xrightarrow{\star} \frac{\pi}{4} \rightarrow |x - \frac{\pi}{4}| \leq \frac{\pi}{4}$

c) $R_2(x) = \frac{1}{2} (2(1) + 2(1)) (\frac{\pi}{4})^2 \approx \frac{\pi^2}{8} \approx 1.2337$

$n=2 \rightarrow R_3(x) = \frac{1}{3!} f^{(3)}(\xi) (x-\frac{\pi}{4})^3 = \frac{1}{6} (2 + 8 \tan^2(\xi) + 6 \tan^4(\xi)) |x-\frac{\pi}{4}|^3$

using results from above, which do not change as the interval remains the same, therefore $|\tan(\xi)| \leq 1, |x-\frac{\pi}{4}| \leq \frac{\pi}{4}$

$R_3 = \frac{1}{6} (2 + 8(1) + 6(1)) (\frac{\pi}{4})^3 = \frac{8\pi^3}{144} = \frac{\pi^3}{18} \approx 1.2919$

$n=3 \rightarrow R_4(x) = \frac{1}{4!} f^{(4)}(\xi) (x-\frac{\pi}{4})^4 = \frac{1}{24} (16 \tan(\xi) + 40 \tan^3(\xi) + 24 \tan^5(\xi)) |x-\frac{\pi}{4}|^4$

Once again, same results for bounds of x and ξ

$R_4 = \frac{1}{24} (16(1) + 40(1) + 24(1)) (\frac{\pi}{4})^4 = \frac{10}{3} (\frac{\pi}{4})^4 = \frac{5\pi^4}{384} \approx 1.2683$

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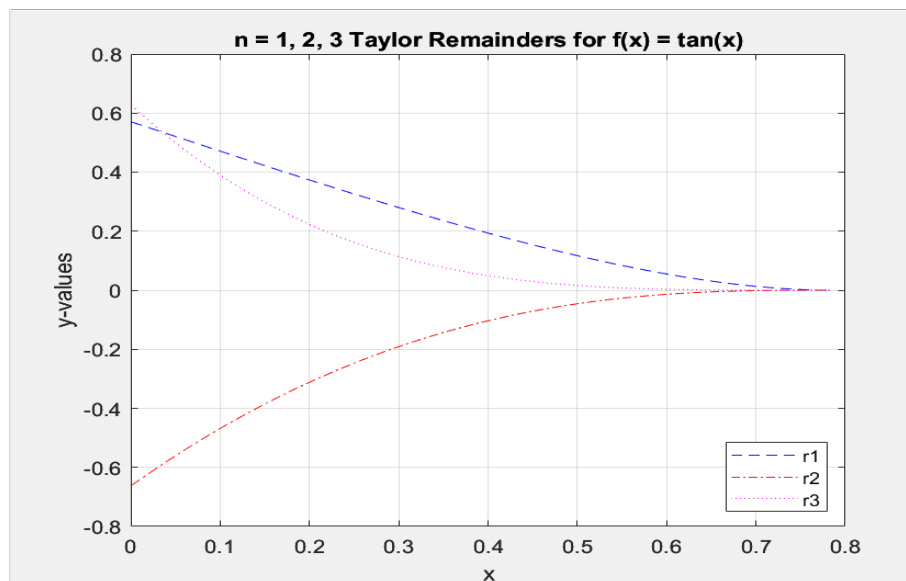
d)

The following code plots the remainders for the first three Taylor approximations of $f(x)$. First we defined a sequence for x on its given interval, then applied component-wise operations to the sequence to “mimic” the behavior of the functions. That is, each polynomial defined below is itself a sequence of values. The remainders are obtained simply by subtracting the actual functional values by the corresponding approximated values for each polynomial. Using the plot function in Matlab/Octave, we were able to visualize the “errors” committed by the Taylor Approximations. An interpretation of the plots follows.

Matlab script (using hold on):

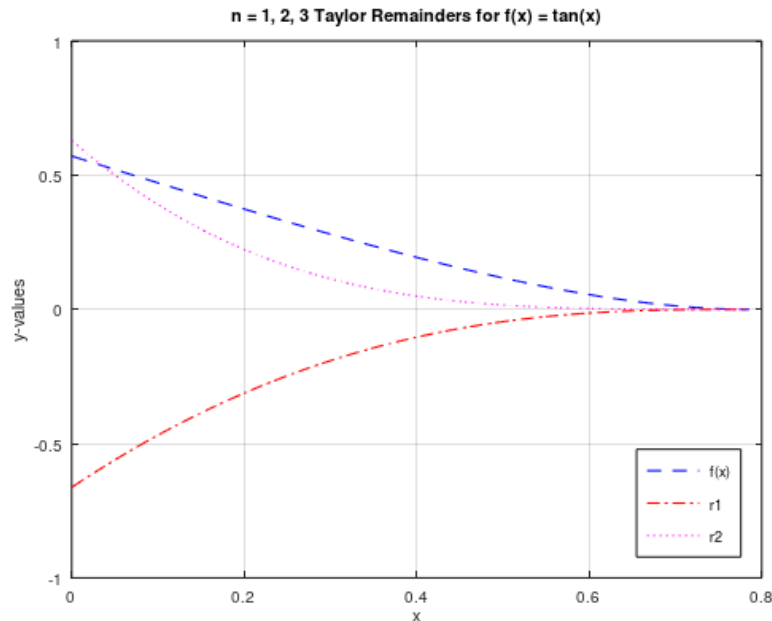
```
x = 0:pi/512:pi/4;  
y = tan(x);  
p1 = 1 + 2*(x-pi/4);  
p2 = 1 + 2*(x-pi/4) + 2*(x-pi/4).^2;  
p3 = 1 + 2*(x-pi/4) + 2*(x-pi/4).^2 + 8/3*(x-pi/4).^3;  
r1 = y-p1;  
r2 = y-p2;  
r3 = y-p3;  
plot(x,r1,'--b')  
grid on;  
xlabel ('x');  
ylabel ('y-values');  
title('n = 1, 2, 3 Taylor Remainders for f(x) = tan(x)');  
hold on  
plot(x,r2,'-.r')  
hold on  
plot(x,r3,':m')  
legend("r1", "r2", "r3", 'Location', 'southeast')  
hold off
```

Graph from Matlab:



Octave Script: (p1, p2, p3 defined in previous octave script included above)

```
r1 = y-p1
r2 = y-p2
r3 = y-p3
plot(x,r1,'--b',x,r2,'-r',x,r3,':m')
grid on;
xlabel ('x');
ylabel ('y-values');
title('n = 1, 2, 3 Taylor Remainders for f(x) = tan(x)');
legend("f(x)", "r1", "r2", "r3", 'Location', 'southeast')
```



The actual remainders are less than their respective theoretical bounds for all values of x on the given interval. It can be seen from the above plot that all three polynomials have the largest remainder at $x = 0$, notably the furthest point away from $\pi/4$ on the given interval. Below is a table of the actual values of these “maximum” bounds, all of which are greater than their corresponding theoretical bounds. These were calculated by substituting 0 for x in each $f(x) - p_n(x)$ equation, then taking the absolute value. Interestingly, both $n=2$ and $n=3$ had larger maximum errors than $n=1$. This is likely due to the power of the expansions being increased along with n , in other words the functions grow faster with larger exponents. p_1 is simply a linear function, whereas p_2 is quadratic, and p_3 is cubic, so it is expected that the latter polynomials will be larger at certain functional values.

n	1	2	3
Maximum Error	0.5707	0.6629	0.6290

“=====”

a)

hw 2 H2

Find smallest integer (positive) such that remainder formula for Taylor approximation guarantees the remainder is less than given tolerance τ

sol $f(x) = e^x$, $a = 1$, $[a, b] = [0, 1]$, $r = 10^{-3} = \frac{1}{10^3} = \frac{1}{1000} = 0.001$

$$Q_{n+1}(x) = \frac{1}{(n+1)!} e^x (x-1)^{n+1}$$

$$\frac{d^1}{dx^n} = c^x \quad \forall n \iff f^{(n)}(x) = f(x) \quad \forall n$$

$$|Q_n(x)| = \left| \frac{1}{(n!)^2} \right| |e^x| |x-1|^{n+1}$$

$$0 \leq \{z\} \rightarrow e^0 z e^0 \rightarrow 1 e^z z e$$

$$0 \leq x \leq 1 \rightarrow -1 \leq x-1 \leq 0 \rightarrow \sqrt{x-1} \leq 0$$

13031-X-31-
↓
1311-X-31-

$\frac{1}{2} \frac{1}{n_{11}}$

$$\frac{1}{(n+1)!} |e| (1) = \frac{1}{(n+1)!} \rightarrow (n+1)! = (n+1)!$$

$$L \rightarrow n=6$$

b)

201

Find lowest positive integer n such that remainder formula for Taylor Approximation guarantees the remainder is less than given tolerance r

$$f(x) = \frac{1}{1-x}, \quad a=0, \quad [\alpha, \beta] = [-\frac{1}{2}, 0], \quad r=10^{-6}$$

$f(x) = (1-x)^{-1}$ * chain rule

$$f'(x) = -(1-x)^{-2} \cdot -1 = (1-x)^{-2}$$

$$f^{(1)}(x) = -2(1-x)^{-3} \cdot -1 = 2(1-x)^{-3}$$

$$f'''(x) = -6(1-x)^{-4} \cdot -1 = 6(1-x)^{-4}$$

$$f^{(n)}(x) = -24(1-x)^{-5} \cdot (-1) = 24(1-x)^{-5}$$

$$f^{(6)}(x) = -120(1-x)^{-6} \cdot -1 = 120(1-x)^{-6}$$

$$\rightarrow f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}} \rightarrow f^{(n+1)}(x) = \frac{(n+1)!}{(1-x)^{n+2}}$$

$$R_{n+1}(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1} \rightarrow \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot \frac{1}{n} x^{n+1}$$

bound terms: ~~free~~

$$-\frac{1}{2} \leq x \leq 0 \rightarrow -\frac{1}{2} \leq x \leq \frac{1}{2} \rightarrow |x| \leq \frac{1}{2}$$

$$\left| \frac{1}{1-z} \right|: \quad -\frac{1}{2} \leq \xi \leq 0 \rightarrow \frac{1}{2} \leq \eta \leq 0 \rightarrow \frac{3}{2} \leq 1-\xi \leq 1$$

Help Me! $\rightarrow \frac{2}{3} \leq \frac{1}{1-x} \leq 1$ finding x $\Rightarrow \frac{1}{3} \leq 1-x \leq 1$

So $\left| \frac{1}{1-s} \right| \leq 1$ then $\left| \frac{1}{1-s} \right|^{1/2} \leq 1$

$\therefore$ then $R_{n+1}(x) = 1 \cdot (1/2)^{n+1} = \frac{1}{2^{n+1}}$

$$\text{let } r = 10^{-6}$$

$$\rightarrow 10^{-6} = \frac{1}{2^{n+1}} \rightarrow \frac{1}{106} = \frac{1}{2^{n+1}} \rightarrow 2^{n+1} = 106$$

$$(n+1) \log(2) = 6 \log(10)$$

$$n+1 = \frac{6 \log(10)}{\log(2)}$$

$$n = \frac{b \log(10)}{\log(2)} - 1 = 18.13$$

Problem 3

HW #2 question/problem 3

quadratic approximation $p_2(x) = f(a) + f'(a)^T(x-a) + \frac{1}{2}(x-a)^T H(a)(x-a)$

$$f(x_1, x_2, x_3) = \frac{1+x_1}{1+x_2} x_3^2, \quad a = (0, 0, 0)^T \quad \text{so } f(a) = 1$$

$$\frac{\partial f}{\partial x_1} = \frac{x_3^2}{1+x_2}, \quad \frac{\partial f}{\partial x_2} = \frac{-x_3^2(1+x_1)}{(1+x_2)^2}, \quad \frac{\partial f}{\partial x_3} = \frac{2x_3(1+x_1)}{(1+x_2)} \quad \rightarrow \quad f'(x) = \begin{bmatrix} \frac{x_3^2}{1+x_2} \\ -\frac{x_3^2(1+x_1)}{(1+x_2)^2} \\ \frac{2x_3(1+x_1)}{1+x_2} \end{bmatrix}, \quad f'(a) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$H(x) = \begin{bmatrix} 0 & -\frac{x_3^2}{(1+x_2)^2} & \frac{2x_3}{1+x_2} \\ -\frac{x_3^2}{(1+x_2)^2} & \frac{2x_3^2(1+x_1)}{(1+x_2)^3} & \frac{-2x_3(1+x_1)}{(1+x_2)^2} \\ \frac{2x_3}{1+x_2} & \frac{-2x_3(1+x_1)}{(1+x_2)^2} & \frac{2(1+x_1)}{1+x_2} \end{bmatrix}, \quad H(a) = \begin{bmatrix} 0 & -1 & 2 \\ -1 & 2 & -2 \\ 2 & -2 & 2 \end{bmatrix}$$

$$H(a) = \begin{bmatrix} x_1 & x_2 & x_3 \\ x_2 & x_1 & x_2 \\ x_3 & x_2 & x_3 \end{bmatrix}$$

$$p_2(x) = 1 + \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3-1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} x_1 & x_2 & x_3-1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 2 \\ -1 & 2 & -2 \\ 2 & -2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3-1 \end{bmatrix}$$

$$= 1 + x_1 - x_2 + 2x_3 - 2 + \frac{1}{2} \begin{bmatrix} x_1 & x_2 & x_3-1 \end{bmatrix} \begin{bmatrix} -x_2+2x_3-2 \\ -x_1+2x_2-2x_3+2 \\ -2x_1-2x_2+2x_3-2 \end{bmatrix}$$

$$= -1 + x_1 - x_2 + 2x_3 + \frac{1}{2} (x_1(-x_2+2x_3-2) + x_2(-x_1+2x_2-2x_3+2) + (x_3-1)(-2x_1-2x_2+2x_3-2))$$

$$= -1 + x_1 - x_2 + 2x_3 + \frac{1}{2} (-x_1x_2 + 2x_1x_3 - 2x_1 - x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_2 + 2x_1x_3 - 2x_1x_2 + 2x_3^2 - 2x_3^2 - 2x_3 + 2)$$

$$= -1 + x_1 - x_2 + 2x_3 + \frac{1}{2} (-4x_1 + 4x_2 - 4x_3 - 2x_1x_2 + 4x_1x_3 - 4x_2x_3 + 2x_2^2 + 2x_3^2 + 2)$$

$$= -1 + x_1 - x_2 + 2x_3 - 2x_1 + 2x_2 - 2x_3 - x_1x_2 + 2x_1x_3 - 2x_2x_3 + x_2^2 + x_3^2 - x_1$$

$$= -x_1 + x_2 - x_1x_2 + 2x_1x_3 - 2x_2x_3 + x_2^2 + x_3^2$$